



An Optimized Wasserstein Deep Convolutional Generative Adversarial Network approach for the classification of COVID-19 and pneumonia

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ABSTRACT

In the context of diagnosing lung disorders like bacterial and viral pneumonia and COVID-19, the challenge of sample scarcity often results in imbalanced datasets, making reliable forecasting difficult. To address this, an Optimized Wasserstein Deep Convolutional Generative Adversarial Network Technique was proposed for the Classification of COVID-19 and Pneumonia (CCP WDCGAN-SOA). The proposed approach utilizes CT scan and X-ray images from two datasets: the COVID-19 Posterior-Anterior Chest Radiography Images Curated Dataset and the COVID QU-Ex Dataset. Due to the imbalance in these datasets, a Label Correlation Guided Borderline Oversampling (LCGBO) method was introduced to balance the classes effectively. Following data balancing, the images undergo pre-processing using Multimodal Hierarchical Graph Collaborative Filtering (MHGCF) for resizing. Subsequently, the processed images are fed into a Wasserstein Deep Convolutional Generative Adversarial Network (WDCGAN) optimized with the Seasons Optimization Algorithm (SOA) to enhance classification accuracy for COVID-19 and pneumonia. The implementation in MATLAB demonstrates that the CCP-WDCGAN-SOA technique significantly outperforms existing methods. Specifically, the proposed approach achieves improvements of 21.5 %, 23 %, and 22.5 % in accuracy, 12.3 %, 17.5 %, and 14 % in recall, and 22.3 %, 27.5 %, and 24 % in specificity compared to DC-CXI-CoviXNet, CPD-CXI-CNN, and ADC-CXI-DFFC Net using the COVID-19 Posterior-Anterior Chest Radiography Images Curated Dataset. Additionally, the proposed method shows gains of 21.52%, 27.05%, and 23.24% in accuracy, 23.71%, 26.45%, and 21.74% in recall, and 28.61%, 22.15%, and 26.44% in specificity over ASC-CXI-LRANet, RCP-MIA-CNN, and AQCD-CR-GAN using the COVID-QU-Ex Dataset.

1. Introduction

Numerous microbiological species, such as viruses, fungi, and bacteria, can cause pneumonia, an infection often contracted. Pneumon, a Greek word meaning “lungs,” is the root of the English term pneumonia. This means that pneumonia and lung damage is related [1,2]. Inflammation of one or both lung parenchymas is the outcome of pneumonia, according to medical terminology. On the other hand, chemical exposure and food aspiration are additional causes of pneumonia [3,4]. Infection-related inflammation results in alveoli in the lung that fill with pus or fluid, leading to pneumonia. This reduces the amount of oxygen (O₂) and carbon dioxide (CO₂) that can be exchanged between the blood and the lungs in ill patients, which makes breathing more difficult [5,6].

A member of the Coronaviridae family, which also causes a novel disease called SARCoV-2, is the source of the highly infectious illness COVID-19. Two members of this family have previously caused a global pandemic: The MERS-CoV and SARS-CoV-1. On December 31, 2019, in Wuhan, in the Hubei province of the Chinese mainland, the first COVID-19 case was reported [7–9]. A worldwide health disaster resulted from the virus's spread from nation to nation and city to city. However, the WHO did not designate it to be a pandemic until March 11, 2020. Respiratory droplets that an infected individual exhales or secretes can spread COVID-19. Coronaviruses invade the alveoli of the lung, an organ involved in the exchange of carbon dioxide and oxygen, and cause pneumonia [10,11]. The symptoms associated with COVID-19 include anorexia, dyspnea, myalgias, severe pneumonia, sputum production,

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organ failure, septic shock, exhaustion, and a dry cough. The SAR-CoV-2 pandemic is concerning because there isn't a vaccination or approved therapy for it as of yet [12].

Differentiating COVID-19 from pneumonia is challenging due to several variables, such as co-infection, overlapping symptoms like fever and cough, variability in imaging features, the potential for disease evolution over time, variability in testing sensitivity, the need for clinical judgment in some situations, resource constraints in others, and the emergence of new variants. Due to these difficulties, a multidisciplinary approach is required to enhance imaging methods, develop more precise and user-friendly diagnostic tools, and improve diagnostic criteria in order to accurately differentiate between the two disorders [13–15]. Deep learning may increase the effectiveness and precision of diagnosis in the healthcare industry when it comes to the classification of pneumonia and COVID-19. But there are several difficulties. Scarcity makes it difficult to obtain labeled data for training, which affects model generalization. Additional challenges include variability and quality problems in medical imaging data, which could jeopardize model performance. Model training is skewed by class imbalance within datasets, which impacts specificity and sensitivity [16,17]. Assuring the model's applicability to a range of demographics is still a challenge. Clinical adoption is hampered by interpretability and explainability issues, and explicit rules are required due to ethical and legal concerns. Deployment is made more difficult by the potential for hostile assaults and resource-intensive training. One of the disadvantages is the absence of uniform datasets, which causes differences in performance assessment and makes it more difficult to compare models [18]. Unbalanced data is a major problem that hinders the model's capacity to generalize, especially for under-represented classes. Restricted generalization outside the particular training datasets is a cause for concern as it may affect the applicability of the system in real life. Deep learning (DL) models are inherently complicated, which creates interpretability challenges that make it challenging for medical practitioners to comprehend and rely on model choices. Over fitting is a prevalent issue that diminishes efficacy in real-world situations, particularly with small datasets and complicated model [19]. Practicality may be limited by computational complexity, which necessitates large resources for training and inference. Model performance is impacted by variability introduced by reliance on image quality [20,21]. This effort was prompted by the shortcomings of the current methods [22]. This study aims to investigate which method is more effective in COVID-19 and Pneumonia Classification.

The novelty of the proposed CCP-WDCGAN-SOA techniques lies in the development of the Label Correlation Guided Borderline Oversampling (LCGBO) technique by offering an innovative solution to the challenge of class imbalance in COVID-19 and pneumonia datasets, ensuring balanced data representation. "Seasons Optimization Algorithm" (SOA) is used to maximize the WDCGAN, improving COVID-19 and Pneumonia classification accuracy and computational efficiency. Overall, the CCP-WDCGAN-SOA model integrates these novel components to outperform existing techniques in the Pneumonia and COVID-19 classifications, demonstrating its efficacy through a thorough evaluation using multiple datasets and performance indicators.

Below is a summary of this research work's principal assistances.

- In this study, An Optimized WDCGAN Method for the COVID-19 Classification and Pneumonia (CCP-WDCGAN-SOA) is proposed.
- Label Correlation Guided Borderline Oversampling (LCGBO) is developed to balance the classes in the imbalance dataset.
- Develop a Multimodal Hierarchical Graph Collaborative Filtering (MHGCF) based preprocessing method for resizing of the collected image.
- WDCGAN is constructed for the COVID-19 Classification and Pneumonia.
- Propose a Seasons Optimization Algorithm (SOA) to optimize the WDCGAN.

- The CCP-WDCGAN-SOA model is used in MATLAB, and several performance metrics are used to evaluate its effectiveness.
- Proposed technique is analyzed using existing techniques like DC-CXI-CoviXNet, CPD-CXI-CNN and ADC-CXI-DFFC Net models from COVID-19 Posterior-Anterior Chest Radiography Images Curated Dataset and ASC-CXI-LRA Net, RCP-MIA-CNN and AQCD-CR-GAN models from COVID-QU-Ex Dataset.

These manuscripts' remaining content is arranged as: Sector 2 examines the survey of relevant literature; Sector 3 outlines a proposed technique; Sector 4 provides commentary to support the Results; and sector 5 wraps up the manuscript.

2. Literature survey

Numerous researches that have been previously published in works have relied on DL for pneumonia and COVID-19 classification. Some of them were mentioned here,

Srivastava et al., [23] have introduced a novel and effective deep learning technique to identify COVID-19 from chest X-ray images. Millions of lives have been lost and great harm to people's mental and physical health has resulted from the coronavirus that started the COVID-19 pandemic. The researchers have tackled the issue of inadequate data for COVID-19 detection by pooling several publically accessible datasets of chest X-rays. This raises concerns since, when two or more datasets were combined; some of the photos may end up being duplicates. For this reason, a curated dataset from an author's work was used in this investigation. This collection includes 3270 normal X-rays, 1281 COVID-19 image, and 1656 chest X-ray pictures with viral pneumonia infection. The dataset has undergone meticulous division and pre-processing to guarantee the absence of duplicate images. It attains higher recall and higher computation time.

Hariri and Avsar, [24] have suggested X-ray technology was a popular and efficient imaging method for diagnosing COVID-19 infections, which have afflicted a lot of individuals globally. Problems could arise from manually examining these X-ray images, particularly in the absence of medical staff. These attempts to develop a CNN-based light weight differential diagnosis model. Using the suggested approach, X-ray image were divided into 4 groups: Infections with bacterial, viral, COVID-19, and healthy pneumonias. The F1-Score, recall, precision, and accuracy to calculate the model's performance was used. The performance of suggested model was contrasted with results from using transfer learning on popular CNN models. It achieves a lower F1-Score and a higher recall.

Liu et al. [25] have investigated Using pictures from chest X-rays, the Deep Features Fusion Classification Network (DFFCNet) was working toward an accurate diagnosis of COVID-19. It was suggested to raise the disease's overall diagnosis accuracy. The MDCM and the deep feature fusion module (DFFM) comprise the two components of the approach. In order to increase classification performance, MDCM employs support vector machines (SVM) as classifiers, while DFFM leverages the advantages of several networks for feature fusion. In the meantime, to enhance the network's capacity for feature extraction, the SA and CA modules were included. Additionally, CXRs are subjected to MDA in order to broaden the diversity of samples. It achieves higher accuracy and computation time.

Yue et al. [26] have presented the "Loop Residual Attention Network for COVID-19 Chest X-Ray Image Automatic Segmentation". Here, a new framework for segmenting COVID-19 CXR images was suggested. In particular, a loop residual module was created that would periodically retrieve feature data during the splicing encoding and decoding process, preventing the loss of intricate semantic information in network computing. In order to improve the location information of feature pixels, a correct position information coding block was also suggested. The establishment of semantic linkages among channels and multi-scale areas was the goal of a hybrid attention module. Combining scale and

location data creates a better feature representation, which lessens the effect of fluctuating infection regions on segmentation performance.

Schraut et al., [27] have presented a multi-output network for medical imaging analysis with a strong classification performance and a U-net improved class activation map. The suggested method demonstrates a multi-output network that employs a U-Net architecture for image segmentation output and an extra CNN module for additional classification output. Class Activation Maps, often known as CAMs, were a way to visualize the feature maps of a CNN that help in classification; in the case of lung diseases, U-net assisted CAM visualization improves the region of interest. The suggested approach uses class activation maps and image segmentation models to extract only the lung region from a chest X-ray's class activation map, producing a representation that increases explain ability and may produce classification outcomes at the same time. This boosts confidence in diagnosing systems driven by AI.

Yoo et al., [28] has investigated the use of generative adversarial networks for the automated quantification of pneumonia caused by the coronavirus in chest radiography. Build an automated method for quantifying COVID-19 pneumonia using chest radiographs and generative adversarial networks (GANs). From each CT scan, full, segmented lung, and pneumonia pixels were used to create anteroposterior virtual radiographs of the chest, lungs, and pneumonia. In order to produce pneumonia images from lung photos and lung pictures from radiographs, two GANs were trained sequentially.

Ravi et al., [29] have presented a DL -based *meta*-classifier that was cost-sensitive for classifying pediatric pneumonia from chest X-rays. The suggested was a deep feature fusion technique for the cost-sensitive large-scale learning of CXR-based pediatric pneumonia categorization using stacked ensemble *meta*-classifier and transfer learning. To determine the relative worth of the various pneumonia classes, additional cost variables were added to transfer learning models, like DenseNet201, Xception, NASNet Mobile, and InceptionResNetV2, based on the degree of class-imbalance during the back propagation learning process. KPCA was then used to lessen the dimensionality of the features that were extracted from InceptionResNetV2, Xception, DenseNet201, and NASNet Mobile's penultimate layer. To categorize the CXR as normal or pneumonia, the stacked ensemble *meta*-classifier was fed the reduced features. Using SVM and random forest for prediction in the 1st stage and logistic regression for classification in the second, a stacked ensemble *meta*-classifier is a 2-step process. It attains a higher F1-Score and lower specificity.

3. Proposed methodology

In this sector, An Optimized WDCGAN Approach for the COVID-19 Classification and Pneumonia (CCP-WDCGAN-SOA) was deliberated. Block schematic of the proposed CCP-WDCGAN-SOA method is illustrated in Fig. 1. It covers such stages as Label Correlation Guided

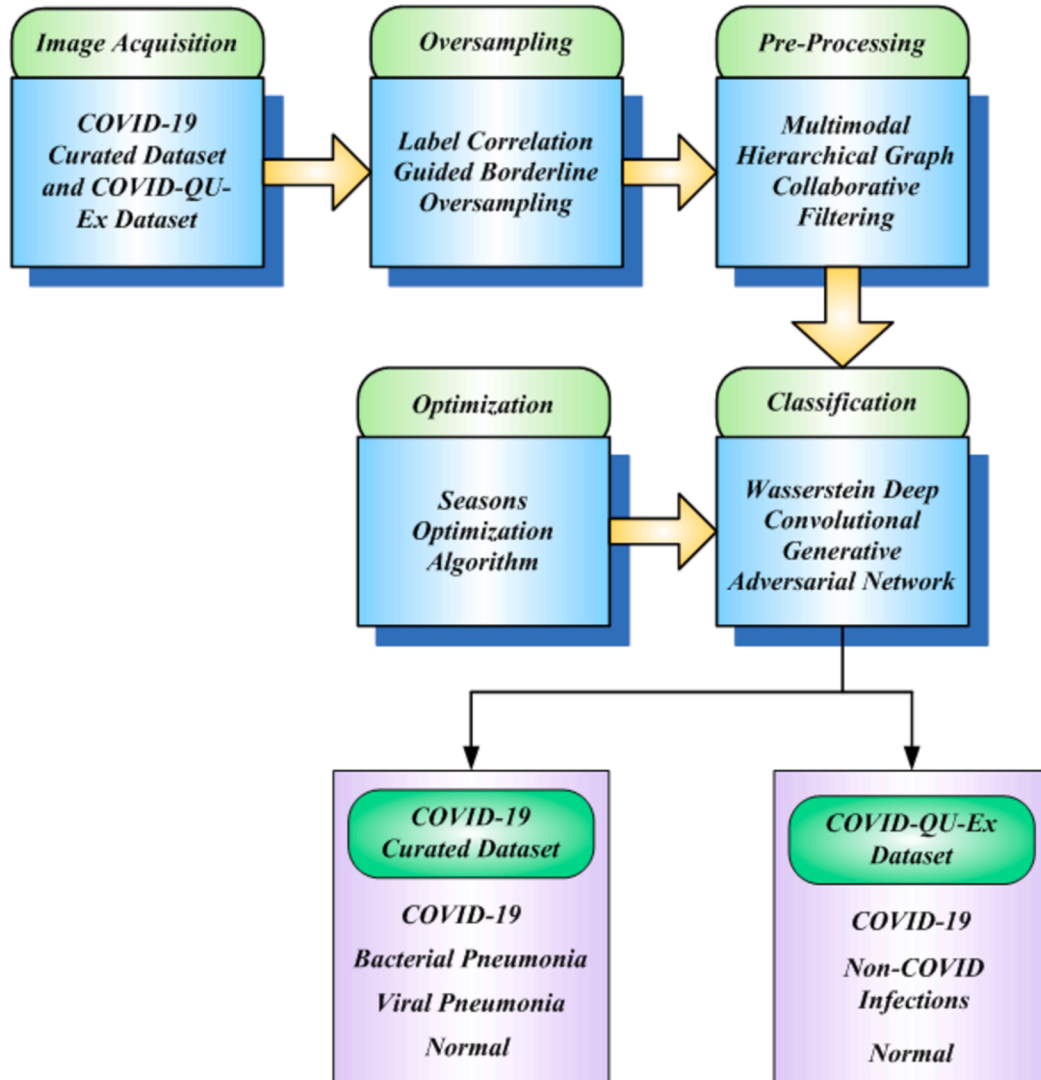


Fig. 1. Block schematic of the proposed CCP-WDCGAN-SOA method.

Borderline Oversampling (LCGBO), Multimodal Hierarchical Graph Collaborative Filtering (MHGCF), Wasserstein Deep Convolutional Generative Adversarial Network(WDCGAN) and Seasons Optimization Algorithm(SOA). Thus, a detailed explanation of every step is given below.

3.1. Image Acquisition

First, the input images are collected from the COVID-19 Curated Dataset [30] and COVID-QU-Ex Dataset [31].

a. COVID-19 Curated Dataset

The first source of images in the COVID-19 Curated Dataset is anterior-posterior chest radiography images. The combined COVID-19 CXR pictures in this carefully chosen collection were created by combining information from 15 publicly accessible datasets, which are listed in the references section. This dataset comprises 1281 X-rays of COVID-19, a total of 3001 bacterial, 1656 viral, and 3270 typical X-ray images were acquired. Table 1 displays the Distribution of COVID-19 Curated Dataset Samples.

b. COVID-QU-Ex Dataset

Researchers at Qatar University created the COVID-QU-Ex dataset, which consists of 33,920 chest X-ray (CXR) images. The collection includes 11,263 non-COVID infections (bacterial or viral pneumonia), 11,956 COVID-19 and 10,701 normal images. The ground-truth lung segmentation masks are included in the entire dataset. This is the biggest lung mask dataset ever produced. Distribution of COVID-QU-Ex Dataset Samples is displayed in Table 2.

3.2. Oversampling using Label Correlation Guided Borderline Oversampling (LCGBO)

In this section, oversampling of the input dataset using Label Correlation Guided Borderline Oversampling (LCGBO) [32] is discussed. There are fewer images in some classes than in others due to the imbalance in the collection of images. Oversampling method LCGBO is used to raise the amount of samples in the minority classes in order to balance the dataset. Equation (1) presents the formulation of learning label correlation as a problem involving sequence prediction.

$$\hat{A}^{(b)} = \sigma(CD + E\hat{A}^{(b-1)}) \quad (1)$$

where, $\hat{A}^{(b)}$ represents the output image, the feature picture is transformed into outer space using the function C , the number of iterations is indicated by b , and into the similar output space as the output of the previous iteration, the output of C using function E . The information provided by the examples on the border is separate even if they are of greater significance.

$$n_p^o(x_i) = \frac{\text{dist}(x_i, s_i^p)}{\text{dist}(x_i, s_i^p)} \quad (2)$$

In this case, the distance between point x and the sample subset s is represented by $\text{dist}(x, s)$, which is the set of x_i 's k -nearest neighbors in R_p ,

Table 1
Distribution of COVID-19 curated dataset samples.

Class	Number of Samples
COVID-19	1281
Bacterial Pneumonia	3001
Viral Pneumonia	1656
Normal	3270

Table 2
Distribution of COVID-QU-Ex dataset samples.

Class	Number of Samples
COVID-19	11,956
Non-COVID infections	11,263
Normal	10,701

and $n_p^o(x_i)$ is the distance factor. Equations (3) and (4) define how many synthetic instances there are for B^O and B^I .

$$G^O = |B^O| \times p \quad (3)$$

$$G^I = |\bar{w}^I > 0| \quad (4)$$

where, p is the sample ratio of B^O , and $|\bar{w}^I > 0|$ is indicated as the amount of cases whose weight is larger than 0. The weights of the instances are used to choose a potential image instance. Since more information-rich candidate examples will more frequently be over-sampled than less information-rich ones, the quality of the newly produced minority instances will be enhanced. After choosing image instances, the next problem is figuring out how to create new instances based on the chosen image instances. Using linear interpolation, a new instance is created from an image instance using the reference instance and the image instance as inputs. The images on the chosen reference instance and the selected image instance must match. As indicated by Tables 3 and 4, LCGBO has finally balanced every class in the input dataset.

3.3. Pre-processing using Multimodal Hierarchical Graph Collaborative Filtering (MHGCF)

In this section, Multimodal Hierarchical Graph Collaborative Filtering (MHGCF) [33] is discussed. The MHGCF gathered information from the COVID-19 Selected Collection of Images in order to do preliminary processing. To resize the image in the preliminary processing phase thus enhancing the quality, the MHGCF is employed. By applying multimodal resizing techniques, MHGCF ensures that images from different modalities are resized consistently. This uniformity is crucial for preserving data integrity and assuring interoperability with DL models. Improve the accuracy of DL models in classifying COVID-19 and Pneumonia by providing high-quality, well-resized images that retain critical diagnostic features. MHGCF integrates images from multiple sources or modalities, creating a unified processing framework. This ensures that images are uniformly processed, despite differences in their original formats or resolutions. It is given in Eq. (5)

$$o_{t_1}^{(k)} = \sum_{h \in M_{t_1} \cup h_1} \frac{|M_h|^\alpha}{|M_{t_1}|^{0.5} |M_h|^{0.5}} o_h^{(k-1)} \quad (5)$$

here, $o_{t_1}^{(k)}$ represent the input image, $k-1$ represent the layer, $o_{t_1}^{(k-1)}$ represent the embedding at layer, $|M_t|$ & $|M_h|$ denotes the size, h denotes the image, α represent the co-efficient and $\frac{|M_h|^\alpha}{|M_{t_1}|^{0.5} |M_h|^{0.5}}$ denotes the size of the image. A graph is constructed where nodes represent image pixels

Table 3
Oversampling of the anterior-posterior chest radiography image curated dataset.

Dataset Class	No of Samples before applying LCGBO	No of Samples after applying LCGBO
Covid-19 X-Rays	1281	3270
Normal X-Rays	3270	3270
Viral-Pneumonia X-Rays	1656	3270
Bacterial-Pneumonia X-Rays	3001	3270

Table 4
Oversampling of COVID-QU-Ex dataset.

Dataset Class	No of Samples before applying LCGBO	No of Samples after applying LCGBO
COVID-19	11,956	11,956
Non-COVID infections (Bacterial or Viral Pneumonia)	11,263	11,956
Normal	10,701	11,956

or patches, and edges represent similarities or relationships. It is given in Eq. (6)

$$p_{d_1}^{(k)} = \sum_{h \in M_{d_1} \cup d_1} \frac{|M_h|^\alpha}{|M_h|^{0.5} |M_{d_1}|^{0.5}} p_h^{(k-1)} \quad (6)$$

here, $p_{d_1}^{(k)}$ represent the image pixels, $k-1$ represent the layer, $p_h^{(k-1)}$ represent the pixel at layer, $|M_h|$ & $|M_{d_1}|$ denotes the size, h denotes the image, α represent the edges. MHGCF identifies and corrects artifacts that introduced during processing, for ensuring that the images used for classification are as clear and accurate as possible. It is given in Eq. (7)

$$g_{l,h_1}^{(1)} = \frac{|M_{h_1}|^\alpha V_l g_{l,h_1}^{(0)}}{|M_{h_1}|} + \sum_{t \in M_{h_1}} \frac{|M_{h_1}|^\alpha h_{l,t}^{(0)}}{|M_{h_1}|^{0.5} |M_t|^{0.5}} \quad (7)$$

here, $g_{l,h_1}^{(1)}$ represents the artifacts, $|M_{h_1}|$ & $|M_t|$ denotes the size, V_l represent the weight transformation matrix, h denotes the image, α represents the edges, l represents the pixels, t_1 denotes the target image and h_1 denotes the identified artifacts. The MHGCF adjusts image contrast to enhance the visibility of important features. This is especially helpful for medical images where minute details can play a crucial role in diagnosis. It is given in Eq. (8)

$$K_1 = \sum_{(t,h,i) \in N} -km\sigma \left(\left(o_t^{(K)} \right)^S \cdot o_h^{(K)} - \left(o_t^{(K)} \right)^S \cdot o_i^{(K)} \right) + \lambda \left(\|O\|_2^2 \right) \quad (8)$$

here, $(t, h, i) \in N$ represents the training image, K_1 objective function, t denotes the image, $\sigma(\cdot)$ represents the sigmoid function and $o_t^{(K)}$ & $o_h^{(K)}$ represents the target image. MHGCF applies coarse-level processing to handle large-scale features and then refines the images through fine-level processing to enhance details. This multi-tiered approach improves the overall quality of the resized images. Thus, Eq. (9) illustrates it.

$$K = K_1 + K_2 + K_3 \quad (9)$$

here, K represent the output image, K_1 is the resized images, K_2 is resized images of the initialized embedding matrix P and K_3 resized images of initialized embedding matrix G . The size in the gathered image from the dataset is resized by MHGCF pre-processing. The pre-processed input was fed into Classification using WDCGAN.

3.4. Classification using Wasserstein Deep Convolutional Generative Adversarial Network (WDCGAN)

In this sector, classification using WDCGAN [34] is discussed. The pre-processed image is classified using WDCGAN. WDCGAN inherits the stability benefits of Wasserstein Generative Adversarial Network, minimizing issues like mode collapse that can occur in traditional Generative Adversarial Network (GAN). This stability is crucial for generating diverse and high-quality synthetic images, which is vital for effective training. The Wasserstein distance provides more informative gradients during training, allowing for better optimization of the generator and critic networks. The combination of deep convolutional layers with the Wasserstein loss function allows WDCGAN to produce

high-fidelity synthetic CXR images that closely resemble real image. WDCGAN aims to improve the diagnostic accuracy in classifying various illnesses, such as pneumonia from chest X-rays and COVID-19. Generator (G) in WDCGAN takes random noise vectors as input and generates synthetic images conditioned on class labels such as COVID-19, pneumonia, or healthy. The discriminator in WDCGAN receives input as a combined vector of actual (real) or generated medical images. The discriminator classifies the real and generated images into the appropriate category (COVID-19, pneumonia, or healthy). It is given in Eq. (10).

$$\min_G \max_D V(D, G) = E_{x \sim P_{data(x)}} [\log D(x|C)] + E_{z \sim P_Z} [\log(1 - D(G(z|C)))] \quad (10)$$

here P_Z specifies the generated image distribution, $P_{data(x)}$ indicates the actual sample's distribution x , $D(G(z|C))$ indicates the likelihood that, given C , the discriminator will determine that the generated sample z is true. $D(x|C)$ is the probability that the discriminator will evaluate the actual sample, x indicates the true under the given C condition, x is the real image. z is represented by a random noise image. The loss function measures how well the classified probabilities align with the true labels. The Wasserstein loss enables the discriminator to provide meaningful feedback to the generator about how "realistic" its outputs are. With an extra gradient penalty, the Wasserstein distance may be interpreted as the WDCGAN's loss function. The Gradient Penalty is to enforce the Lipschitz constraint required for WDCGAN. This loss function is defined using Eq. (11)

$$L = E_{z \sim P_g} [D(z)] - E_{x \sim P_r} [D(x)] + \lambda_{z \sim P_g} E \left[\left(\|\nabla_{\hat{z}} D(\hat{z})\|_2 - 1 \right)^2 \right] \quad (11)$$

Eq. (11) represents the Wasserstein distance between the actual samples x and the generator output z . P_g and P_r , are two probability distributions. It is necessary to decrease this WDCGAN loss function when learning. The gradient penalty refers to the penalty applied to the gradient norm on the samples where P_Z , represents the distribution acquired from uniform sampling along a linear segment connecting the produced and actual distributions, P_g and P_r . F_j in the WDCGAN classification model is taught using the X usage data. The GAN task of differentiating between real and generated images is combined with the trained discriminator to conduct direct classification of X-ray images into COVID-19, pneumonia, or healthy categories. Formula Eq. (12) is used to classify COVID-19 and pneumonia based on the input images.

$$Y_j = F_j(w_j, X_j) \quad (12)$$

here, w_j and Y_j are the factors to be learned and the given label of the j -th characteristics, correspondingly and F_j is the classified image. Finally, the WDCGAN has classified the COVID-19 and Pneumonia from the pre-processed images. Convenience and pertinence make the AI-depend optimization strategy a WDCGAN classifier. In this study, the WDCGAN ideal parameters F_j are optimized with the use of SOA. Here, the weight and bias factor of the WDCGAN are turned using SOA.

3.5. Optimization using Seasons Optimization Algorithm (SOA)

Using a proposed Seasons Optimization Algorithm (SOA) [35], the weight factor F_j of the proposed WDCGAN is optimized. Seasonal variations could help the model become less prone to over fitting to the training set by making it more flexible to fresh data. The method may be able to lower the incidence of incorrect diagnoses by taking seasonal fluctuations into account, improving patient care. SOA mimics natural adaptation and evolution, with populations responding to seasonal environmental variations. It iteratively updates candidate solutions, taking into account individual fitness and the effects of seasons such as spring for exploration and fall for exploitation. This method enables SOA to efficiently transverse search spaces and converges on optimal

solutions, making it appropriate for a broad variety of optimization tasks.

3.5.1. Stepwise procedure for seasons optimization algorithm

Here, a step-by-step procedure for employing SOA to acquire suitable WDCGAN values is shown. In order to improve the optimal WDCGAN parameters, a population with a uniform distribution is created. The entire procedure is shown here,

Step 1: Initialization

This initial population forms the basis for the algorithm's exploration and exploitation stages. As the algorithm iterates, solutions change in response to seasonal conditions.

Then the initialization process is given in Eq. (13)

$$Z = \begin{bmatrix} Z_{1,1} & \dots & Z_{1,j} & \dots & Z_{1,F} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ Z_{i,1} & \dots & Z_{i,j} & \dots & Z_{i,F} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ Z_{M,1} & \dots & Z_{M,j} & \dots & Z_{M,F} \end{bmatrix}_{M \times F} \quad (13)$$

where, (G) represents the candidate solution of SOA, (Z_i) denotes i^{th} candidate solution of SOA, (F) represents decision variables of a given problem, (M) denotes SOA members, (z_{ij}) indicates value of j^{th} variable specific by i^{th} candidate solution.

Step 2: Random Generation

The settings entered were chosen at random. Based on a clear hyperparameter condition, the ideal fitness values were chosen.

Step 3: Fitness Function

Generate an arbitrary result using the initialization settings. Equation

(14) is used to assess the fitness function.

$$\text{Fitness function} = \text{Optimization } [F_j] \quad (14)$$

Step 4: Exploration Phase

At this stage, exploration and exploitation must be balanced to ensure effective exploration and the utilization of potential locations for optimization. Then, the exploration phase is given in equation (15)

$$Q(r+1) = \text{Best}_g \times S + ((\text{Maximum} - \text{Minimum}) \times S + \text{Minimum}) \times L(F) \quad (15)$$

here, $(Q(r+1))$ represents the seasonal environmental variations, (Best_g) denotes the current search space selected by their last search, (S) and represents the random number takes a value among 0, 1. $(L(F))$ represents Levy flight,

Step 5: Exploitation Phase for Optimizing F_j

As opposed to the fall, the exploitation phase speeds up the process of identifying potential areas inside the solution space. This phase builds on the insights gained from exploration, refining possible solutions until they converge on the ideal option. Its purpose is to maximize efficiency while achieving convergence to the global optimum. Thus, the exploitation phase is given in Eq. (16)

$$F_j = \text{Best}_g \times l(t) + \text{Mean_Info} - F_j + 1 \times l(t) \quad (16)$$

where, $F_j + 1$ represents the global optimum, $(l(t))$ represents the position-change-controlling parameter, (Mean_info) Represents mean, F_j denotes the position of giant trevally (i) at time (r) .

Step 6: Termination Criteria

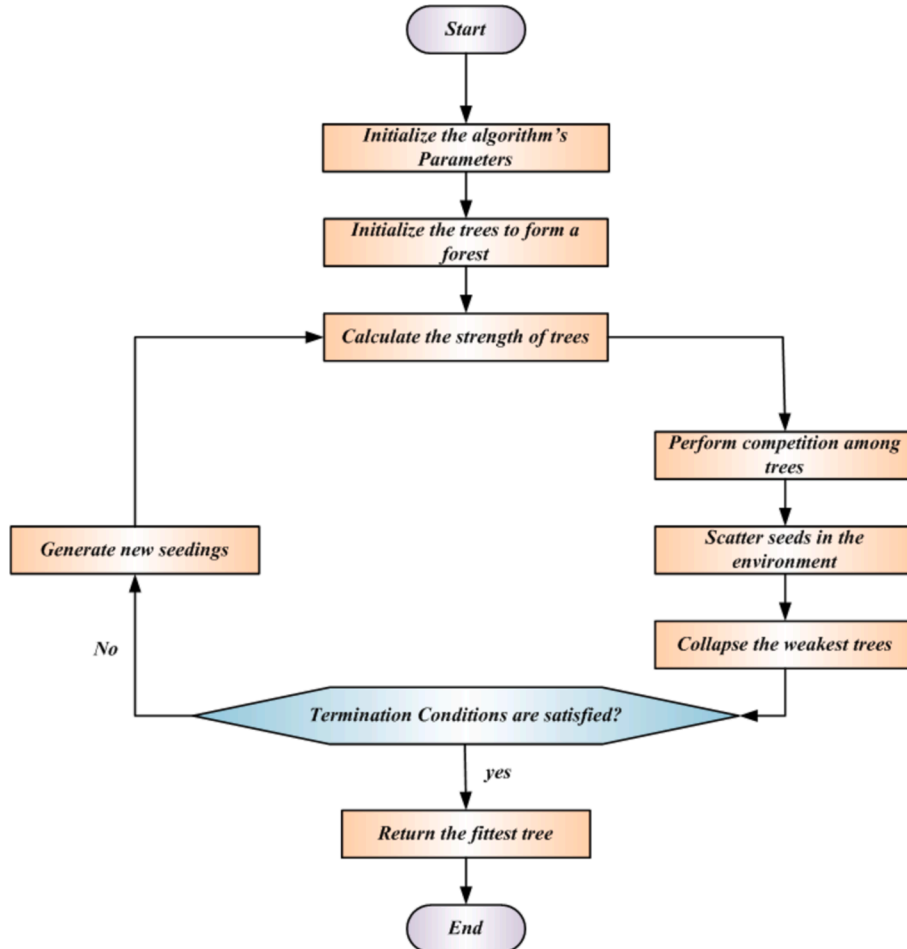


Fig. 2. Flowchart of Seasons Optimization Algorithm (SOA).

Step 3 should be repeated iteratively until the halting requirement $Z = Z + 1$ is satisfied. The weight parameter F_j values of the generator from WDCGAN are improved with the assistance of SOA. Then WDCGAN classified the pneumonia and COVID-19 with reduced computational time and increased precision. Fig. 2 displays the Flow-chart of SOA.

4. Result with Discussion

This section discusses the recommended method's experimental results. Next, utilizing the previously specified performance metrics, the

proposed technique is simulated in MATLAB. In MATLAB, the proposed CCP-WDCGAN-SOA approach is used. The output of the proposed CCP-WDCGAN-SOA strategy is compared to existing systems. The output results of proposed CCP-WDCGAN-SOA approach are displayed in Fig. 3.

4.1. Performance measures

Selecting the best classifier requires taking this critical step. Numerous measures are used to assess performance, like specificity, recall, F1-score, accuracy, and precision as well as calculation time. To scale it, the performance measure is established. In order to increase the

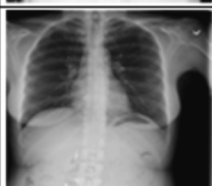
<i>Dataset</i>	<i>Input Image</i>	<i>Pre-Processed Image</i>	<i>Output</i>
<i>Covid 19 Curated Dataset</i>			<i>Covid 19</i>
			<i>Bacterial Pneumonia</i>
			<i>Viral Pneumonia</i>
			<i>Normal</i>
<i>Covid QU-Ex Dataset</i>			<i>Covid 19</i>
			<i>Non-Covid Infections</i>
			<i>Normal</i>

Fig. 3. Output result of proposed CCP-WDCGAN-SOA approach.

performance metric's scale Samples of True Negative, True Positive, False Negative and False Positive are required.

- True Negative (TN): Presents the count of samples which are properlyforecast as negative.
- True Positive (TP): Presents the count of samples values which are properlyforecast as positive.
- False Positive (FP): Presents the count of positive samples which are imperfectlyforecast as positive.
- False Negative (FN): Presents the count of samples which are imperfectlyforecast as negative

4.1.1. Accuracy

Eq. (17) provides the accuracy, which quantifies the percentage of samples (both negative and positive) in addition to the entirecount of samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

4.1.2. Precision

Accuracy of the positive forecast produced by an ML model is a measure of its success, in addition to precision.The precision may be computed as follows: entire count number of positive predictions divided by the entire countof genuine positives, is provided by equation (18).

$$Precision = \frac{TP}{TP + FP} \quad (18)$$

4.1.3. Recall

The recall, or true positive rate, of a machine learning model is the proportion of data samples that the model correctly classifies as being in a class of interest.The following Eq. (19) is used to measure it.

$$Recall = \frac{TP}{TP + FN} \quad (19)$$

4.1.4. Specificity

The capacity of a model to forecast true negatives for each category is measured by its specificity. You may use these metrics with any category model.

Eq. (20) provides it.

$$Specificity = \frac{TN}{TN + FP} \quad (20)$$

4.1.5. F1-Score

One metric utilized to evaluate an ML model's performance is the F-score. Equation (21), which provides a single score, combines recall and accuracy.

$$F1 - Score = 2 * \frac{precision * recall}{precision + recall} \quad (21)$$

4.2. Performance analysis of CCP-WDCGAN-SOA on COVID-19 curated dataset

Figs. 4 to 7 or portrays the simulation outcomes of proposed CCP-WDCGAN-SOA technique. Then, the proposed CCP-WDCGAN-SOA technique is likened with existing DC-CXI-CoviXNet, CPD-CXI-CNN and ADC-CXI-DFFCNetmethods respectively.

Fig. 4 illustrates Accuracy analyses. The proposed CCP-WDCGAN-SOA method attains 29 %, 27.5 % and 28 % higher accuracy for COVID-19; 24 %, 28.5 % and 26 % higher accuracy for Bacterial Pneumonia; 21.5 %, 23 %, and 22.5 % higher accuracy for Viral Pneumonia and 19.5 %, 24 % and 23 % greater accuracy for Normal

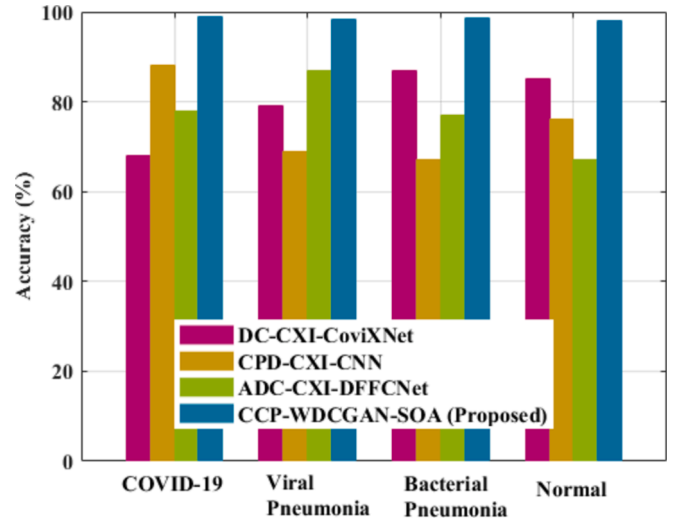


Fig. 4. Performance analyses of accuracy.

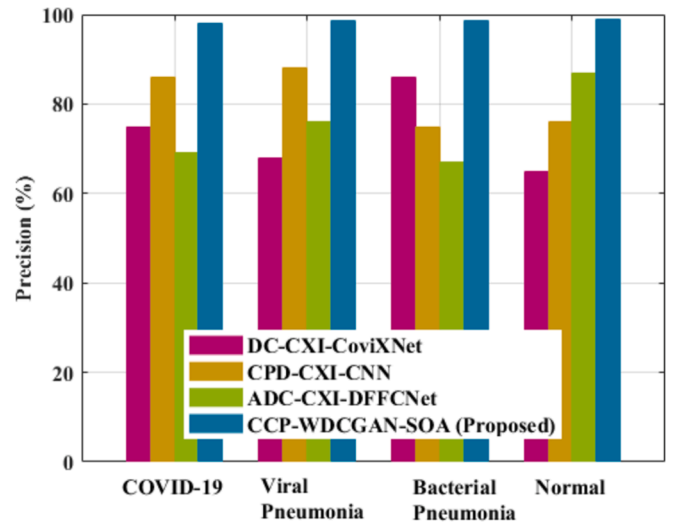


Fig. 5. Performance analyses of precision.

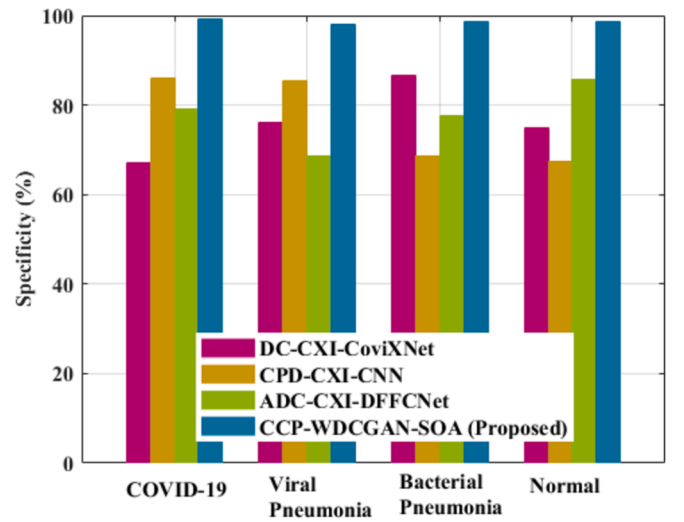


Fig. 6. Performance analyses of specificity.

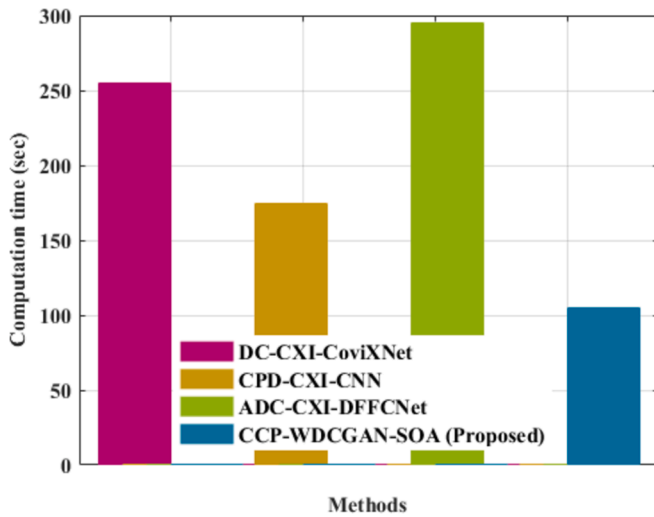


Fig. 7. Performance analyses of computation time.

estimated to the existing DC-CXI-CoviXNet, CPD-CXI-CNN and ADC-CXI-DFFCNet models respectively.

Fig. 5 displays Precision analyses. The proposed CCP-WDCGAN-SOA technique attains 23 %, 21.5 % and 20 % higher precision for COVID-19; 14 %, 18.5 % and 16 % higher precision for Bacterial Pneumonia; 25 %, 21.5 %, and 22.5 % higher precision for Viral Pneumonia and 17.2 %, 20 % and 28.5 % greater precision for Normal estimated to the existing DC-CXI-CoviXNet, CPD-CXI-CNN and ADC-CXI-DFFCNet models

respectively.

Fig. 6 displays Specificity analyses. The proposed CCP-WDCGAN-SOA method attains 24.12 %, 27.05 % and 25.3 % higher Specificity for COVID-19; 22.3 %, 27.5 % and 24 % higher Specificity for Bacterial Pneumonia; 15.3 %, 17.5 %, and 21.04 % higher Specificity for Viral Pneumonia and 28.2 %, 27.06 % and 17.87 % greater Specificity for Normal estimated to the existing DC-CXI-CoviXNet, CPD-CXI-CNN and ADC-CXI-DFFCNet models correspondingly.

Fig. 7 displays computation time analyses. The proposed CCP-WDCGAN-SOA method attains 21.3 %, 25.1 % and 22.3 % lesser computation time estimated to the existing technique such as DC-CXI-CoviXNet, CPD-CXI-CNN and ADC-CXI-DFFCNet models respectively.

Confusion Matrix of COVID-19 Curated Datasets is displayed in Fig. 8. A proposed CCP-WDCGAN-SOA model's performance is assessed using a confusion matrix on a COVID-19 curated dataset that has 4 classes: Normal, Viral, COVID-19, and Bacterial Pneumonia. Each category's correctly classified instances are represented by the diagonal values in the matrix. Specifically, the proposed CCP-WDCGAN-SOA model correctly identified 957 instances of COVID-19, 959 cases of bacterial pneumonia, 958 cases of viral pneumonia, and 959 cases of normal pneumonia. The off-diagonal data showed that one case of COVID-19 was mistakenly diagnosed as viral pneumonia and one instance of normal pneumonia was mistakenly diagnosed as bacterial pneumonia. In all categories, the proposed CCP-WDCGAN-SOA model shows very few misclassifications and a high degree of accuracy.

Table 5 shows the Comparative analysis of existing models DC-CXI-CoviXNet, CPD-CXI-CNN, ADC-CXI-DFFCNet and the proposed CCP-WDCGAN-SOA on COVID-19 curated dataset. The models were tested on four classifications like Bacterial Pneumonia, Viral Pneumonia,

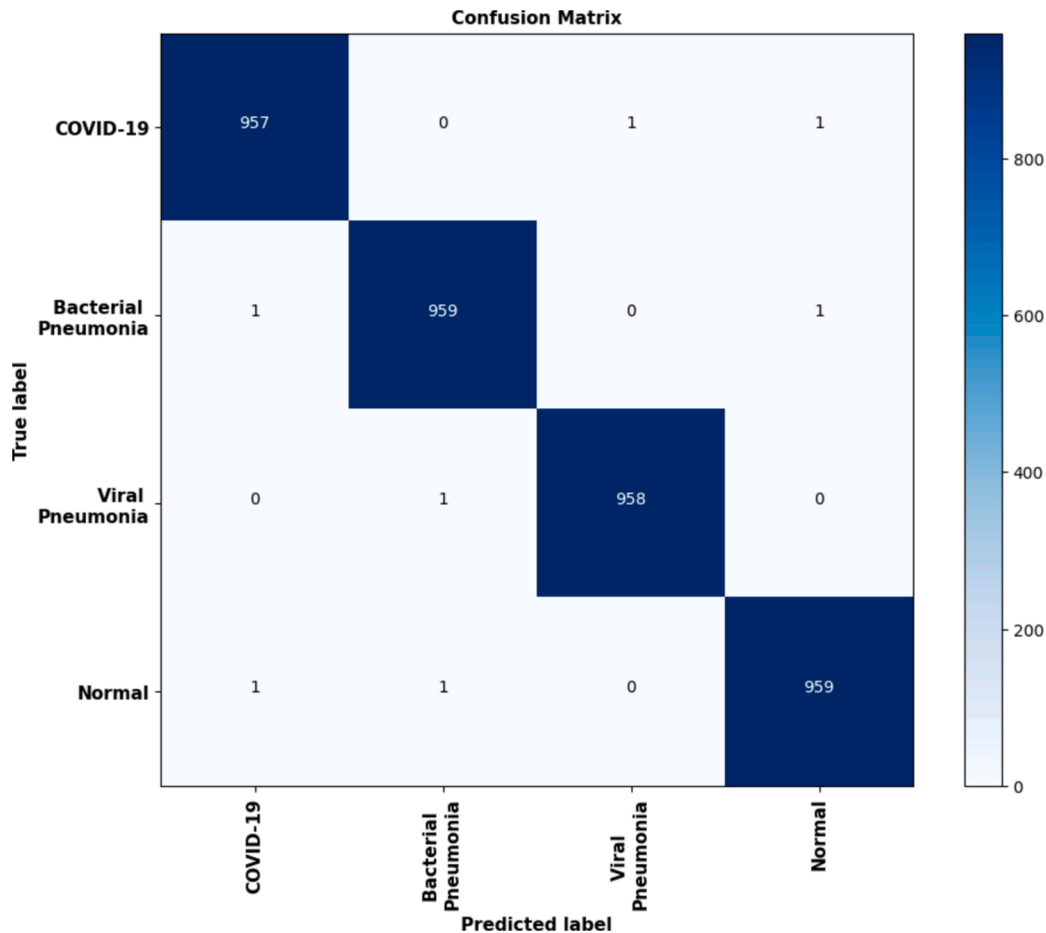


Fig. 8. Confusion matrix of COVID-19 curated dataset.

Table 5

Performance of CCP-WDCGAN-SOA on COVID-19 curated dataset.

Dataset Class		DC-CXI-CoviXNet	CPD-CXI-CNN	ADC-CXI-DFFCNet	CCP-WDCGAN-SOA (Proposed)
COVID-19	Recall (%)	74.35	66.27	66.74	96.25
	F1-Score (%)	65.42	77.56	77.68	97.52
Viral Pneumonia	Recall (%)	63.75	67.06	67.24	97.05
	F1-Score (%)	68.97	66.85	66.51	96.87
Bacterial Pneumonia	Recall (%)	78.65	69.14	79.15	99.15
	F1-Score (%)	74.45	77.32	77.42	97.35
Normal	Recall (%)	69.27	78.01	68.45	98.07
	F1-Score (%)	74.36	66.90	76.12	96.96

COVID-19, and Normal using Recall (%) and F1-Score (%) as evaluation metrics. In all classifications, the proposed CCP-WDCGAN-SOA model performs well contrasted to other models. The aforementioned results demonstrated the availability and accuracy of proposed CCP-WDCGAN-SOA model as a valid standard for diagnosis compared to related models that have been already utilized.

4.3. Performance analysis of CCP-WDCGAN-SOA on COVID-QU-Ex dataset

Figs. 9 to 12 portrays the simulation outcomes of proposed CCP-WDCGAN-SOA technique. Then, the proposed CCP-WDCGAN-SOA technique is likened with existing ASC-CXI-LRANet, RCP-MIA-CNN and AQCD-CR-GAN methods respectively.

Fig. 9 illustrates Accuracy analyses. The proposed CCP-WDCGAN-SOA technique attains 24.21 %, 20.31 % and 25.36 % higher accuracy for COVID-19; 21.52 %, 27.05 % and 23.24 % higher accuracy for Non-Covid Infections and 29.74 %, 25.62 % and 24.31 % greater accuracy for Normal estimated to the existing ASC-CXI-LRANet, RCP-MIA-CNN and AQCD-CR-GAN models respectively.

Fig. 10 illustrates precisionanalyses. The proposed CCP-WDCGAN-SOA method attains 23.71 %, 26.45 % and 21.74 % higher precision for COVID-19; 27.02 %, 26.65 % and 20.85 % higher precision for Non-Covid Infections and 24.14 %, 26.02 % and 27.31 % greater precision for Normal estimated to the existing ASC-CXI-LRANet, RCP-MIA-CNN and AQCD-CR-GAN models respectively.

Fig. 11 illustrates Specificity analyses. The proposed CCP-WDCGAN-SOA method attains 28.61 %, 22.15 % and 26.44 % higher Specificity for COVID-19; 21.12 %, 20.06 % and 28.15 % higher Specificity for Non-Covid Infections and 25.54 %, 21.12 % and 22.21 % greater Specificity for Normal estimated to the existing ASC-CXI-LRANet, RCP-MIA-CNN and AQCD-CR-GAN models respectively.

Fig. 12 illustrates Computational Time analyses. The proposed CCP-WDCGAN-SOA method attains 23.64 %, 20.77 % and 24.14 % lower

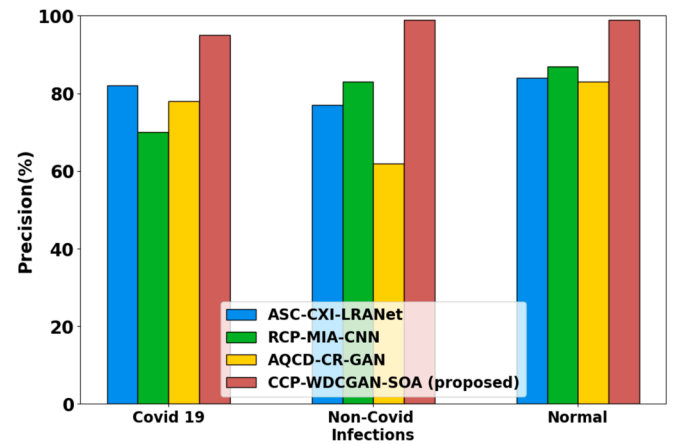


Fig. 10. Performance analyses of precision.

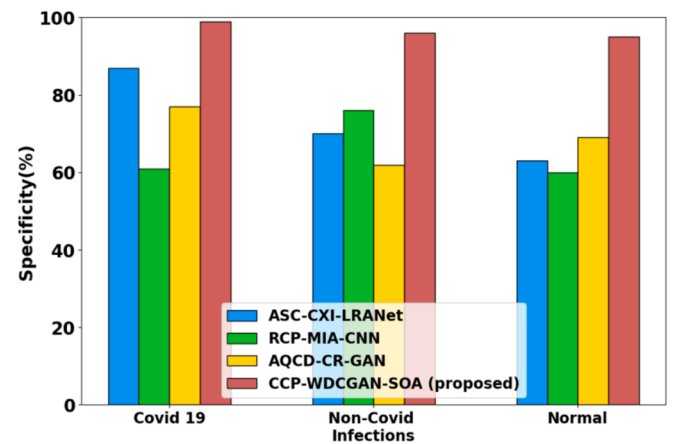


Fig. 11. Performance analyses of specificity.

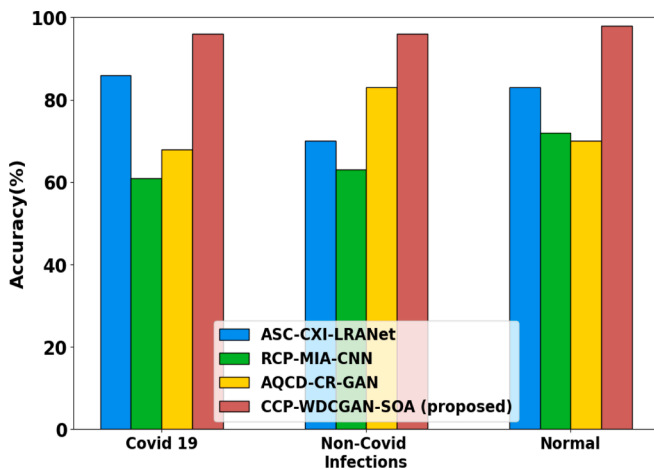


Fig. 9. Performance analyses of accuracy.

Computational Time estimated to the existing ASC-CXI-LRANet, RCP-MIA-CNN and AQCD-CR-GAN models respectively.

Fig. 13 displays the Confusion Matrix of COVID-QU-Ex dataset. The confusion matrix denotes the proposed CCP-WDCGAN-SOA model performance of COVID-QU-Ex dataset with 3 classes namely Non-COVID Infections, COVID-19, and Normal. The values in the diagonal are correct classifications of examples for each class. Theproposed CCP-WDCGAN-SOA model correctly predicted 5,687 COVID-19 instances, 5,686 Non-COVID Infections cases and 5,687 Normal cases. Misclassifications are minimal, with only one instance of COVID-19 being misclassified as Normal, one instance of Non-COVID Infections as COVID-19, and one instance of Normal as Non-COVID Infections. This matrix proves that theproposed CCP-WDCGAN-SOA model maintains high accuracy in all categories by having a few numbers of misclassifications.

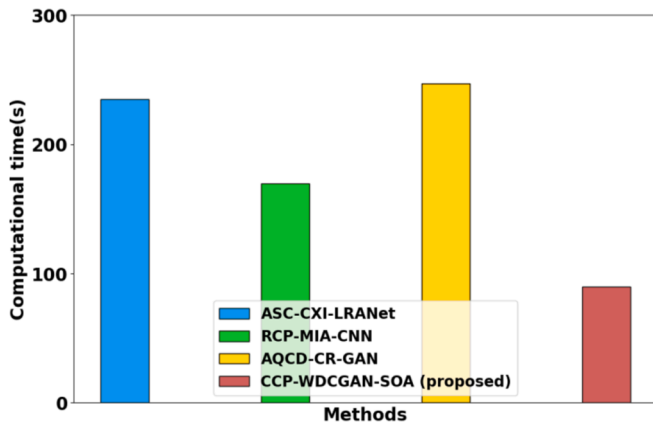


Fig. 12. Performance analyses of computational time.

Table 6 shows the Comparative analysis of existing models ASC-CXI-LRANet, RCP-MIA-CNN and AQCD-CR-GAN and the proposed CCP-WDCGAN-SOA on COVID-QU-Ex dataset. The models were tested on three classifications such as Non-Covid Infections, COVID-19, and Normal using Recall (%) and F1-Score (%) as evaluation metrics. In all classifications, the proposed CCP-WDCGAN-SOA model performs well compared to other models.

The Fig. 14 displays the loss curves for the Generator (gen) and Critic (crit_real and crit_fake) during WDCGAN training. In WDCGAN, both the generator and critic (discriminator) networks compete in a minimax game. Whereas the critic seeks to distinguish between created (false) and actual images, the generator seeks to create realistic synthetic visuals. crit_real (blue line) represents the critic's loss when processing real images, while crit_fake (orange line) shows the critic's loss for

generated images. gen (green line) tracks the generator's loss during training as it strives to create images that closely resemble real ones. Early around 0–100 epochs, the crit_real loss (blue) rapidly decreases from approximately 0 to –20, while the crit_fake loss (orange) increases from 0 to around 10. The generator loss (green) starts near 0 and initially fluctuates, rising to about 10 by 200 epochs. As training progresses 200–600 epochs, the generator loss continues to increase steadily, reaching around 20, while the critic losses stabilize, with critreal around –10 to –5 and crit_fake around 5 to 10. By the end of 600–1000 epochs, the generator loss remains elevated, hovering around 20–25, and the critic losses maintain a stable gap, indicating that both networks have reached a balanced training phase.

4.4. Ablation study

The proposed CCP-WDCGAN-SOA's ablation analysis is performed both with and without each component to ascertain its level of significance.

Table 7 displays the ablation study of the proposed CCP-WDCGAN-SOA method, here the performance of MHGCF, LCGBO, WDCGAN, SOA and the proposed CCP-WDCGAN-SOA method has been evaluated across two various datasets namely COVID-19 Curated dataset and COVID-QU-Ex dataset. The study considered various performance metrics, including Specificity, Accuracy, F1-Score, Recall, Computational Time and Precision. The findings indicate that the proposed CCP-WDCGAN-SOA method outperforms MHGCF, LCGBO, WDCGAN and SOA in terms of accuracy, with significantly higher values for COVID-QU-Ex dataset than COVID-19 Curated dataset. Additionally, it produces substantial improvements in f1-score and specificity, indicating that the proposed approach improves categorization. Furthermore, the proposed CCP-WDCGAN-SOA technique is both computationally

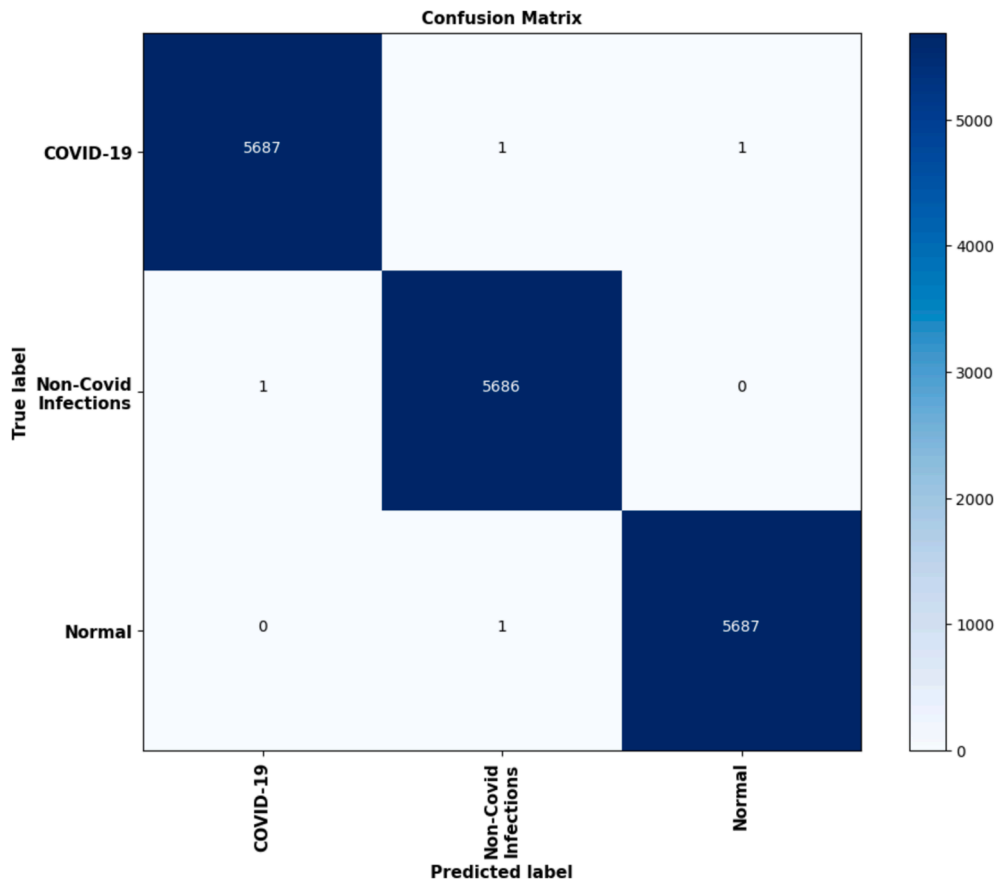
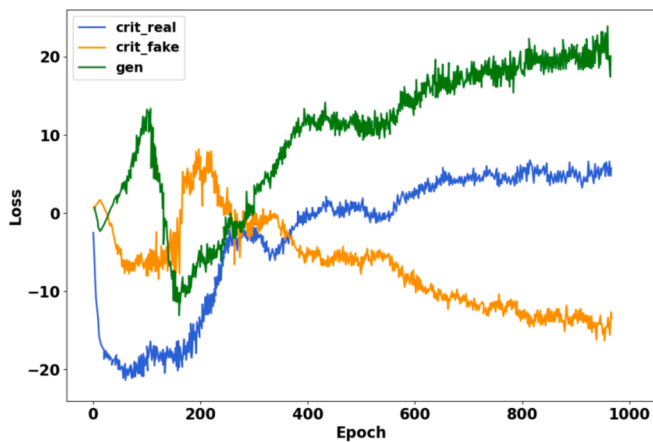


Fig. 13. Confusion matrix of COVID-QU-Ex dataset.

Table 6

Performance of CCP-WDCGAN-SOA on COVID-QU-Ex dataset.

Dataset Class		ASC-CXI-LRANet	RCP-MIA-CNN	AQCD-CR-GAN	CCP-WDCGAN-SOA (Proposed)
COVID-19	Recall (%)	74.21	76.24	76.34	97.74
	F1-Score (%)	65.12	67.54	67.28	98.32
Non-Covid Infections	Recall (%)	63.35	77.01	77.44	98.55
	F1-Score (%)	68.07	76.81	76.41	97.57
Normal	Recall (%)	78.15	79.13	69.45	96.35
	F1-Score (%)	74.05	67.30	67.32	95.05

**Fig. 14.** Loss graph of Wasserstein Deep Convolutional Generative Adversarial Network (WDCGAN).

economical and more accurate due to its notable reduction in processing time. These results highlight the utility of the MHGCF, LCGBO, WDCGAN, and SOA fusion in enhancing pneumonia and COVID-19 classification in the difficult COVID-19 Curated dataset and COVID-QU-Ex dataset.

4.5. Discussion

A novel An Optimized WDCGAN Approach for the COVID-19 and Pneumonia Arrangements is developed in this paper. The Selected

Collection of COVID-19 Front-Anterior Chest Radiography images and COVID-QU-Ex Dataset is used as the basis for the classification of different COVID-19 and pneumonia categories. It brought up the issue of their classification, which has grown more difficult as a result of class imbalance in dataset for classifying. The oversampling technique Label Correlation Guided Borderline Oversampling (LCGBO) is used to balance the classes in the dataset. Focus was placed on classifying COVID-19 and pneumonia with the CX-images in order to address this classification problem. The proposed CCP-WDCGAN-SOA method not only achieves significantly higher accuracy, recall, and specificity compared to existing techniques but also demonstrates efficient computational performance suitable for clinical applications. However, enhancing diagnostic accuracy necessitates a strong focus on interpretability to gain trust among healthcare professionals. Incorporating techniques like Class Activation Maps (CAM) and Gradient-weighted Class Activation Mapping (Grad-CAM) will provide insights into the model's decision-making process by highlighting influential image regions, while attention mechanisms can improve focus on critical features. Additionally, utilizing explainability tools like SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) can clarify how individual features contribute to outcomes, ensuring the model aligns with clinical reasoning. Collaborating with medical professionals throughout development will foster a deeper understanding of clinical contexts and facilitate feedback, ultimately enhancing the model's integration into clinical workflows and supporting improved patient outcomes through greater transparency and practical utility in medical diagnostics.

5. Conclusion

In this section, An Optimized Wasserstein Deep Convolutional Generative Adversarial Network Approach for the COVID-19 and Pneumonia Classification (CCP-WDCGAN-SOA) is successfully implemented. In MATLAB, the proposed CCP-WDCGAN-SOA method is put into practice. The performance of the proposed CCP-WDCGAN-SOA method contains 25 %, 21.5 %, and 22.5 % higher precision; 27.6 %, 24.8 %, and 23.2 % higher F1-Score and 21.3 %, 25.1 % and 22.3 % lesser computation timewhen analyzed with DC-CXI-CoviXNet, CPD-CXI-CNN and ADC-CXI-DFFCNet methods for COVID-19 curated dataset and 24.14 %, 26.02 % and 27.31 % higher precision, 25.54 %, 21.12 % and 22.21 % higher Specificity and 23.64 %, 20.77 % and 24.14 % lower Computational Timewhen analyzed with ASC-CXI-LRANet, RCP-MIA-CNN and AQCD-CR-GAN methods for COVID-QU-Ex Dataset. Future research should concentrate on improving the interpretability of the CCP-WDCGAN-SOA method by integrating visualizations like saliency maps and confidence scores. Enhancing these features will help align the model's predictions with clinical reasoning, ensuring that the method supports informed and transparent decision-making in medical practice.

Table 7

Ablation Study of proposed CCP-WDCGAN-SOA method.

Performance Metrics	Dataset	Methods				
		Without MHGCF	Without LCGBO	Without WDCGAN	Without SOA	CCP-WDCGAN-SOA (Proposed)
Accuracy (%)	COVID-19 Curated	82.23	84.71	88.04	80.93	99.86
	COVID-QU-Ex	84.31	80.11	84.32	74.35	99.92
Precision (%)	COVID-19 Curated	80.54	74.31	84.02	71.21	99.74
	COVID-QU-Ex	84.44	70.21	80.32	81.71	99.83
Recall (%)	COVID-19 Curated	81.23	74.36	85.19	77.51	99.65
	COVID-QU-Ex	80.13	71.85	80.75	73.11	99.95
Specificity (%)	COVID-19 Curated	80.25	84.36	81.25	81.21	99.91
	COVID-QU-Ex	78.05	82.48	74.05	75.31	99.96
F1-Score (%)	COVID-19 Curated	74.35	81.18	71.35	70.11	96.32
	COVID-QU-Ex	74.35	72.08	71.35	69.21	97.03
Computational Time (Sec)	COVID-19 Curated	147	151	185	163	101
	COVID-QU-Ex	135	149	179	158	92

Availability of supporting data

This research is exempt from the data sharing policy because it did not use any newly created or analyzed data.

CRedit authorship contribution statement

A.B. Rajendra: Writing – original draft. **B.S. Jayasri:** Supervision. **S. Ramya:** Supervision. **Shruthi Jagadish:** Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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